

Big Data

Chapter 7

NoSQL and NewSQL Benchmarking

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BENCHMARKS
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Agenda

- Survey of existing NoSQL benchmarks
- Properties for a relevant NoSQL benchmark
- The state of NoSQL benchmarks
  - Key-value store benchmarks
  - Document store benchmarks
  - Column store benchmarks
  - Graph store benchmarks
- Examples of practical works
- Conclusion.

Benchmarking

- A plethora of NoSQL technologies has emerged as currently over 300 NoSQL technologies are in existence.

- Due to the large heterogeneity in database API's, terminology, and data models, it has become increasingly difficult to compare these technologies in terms of their features, database functionality, and especially performance.

- The Yahoo! Cloud Serving Benchmark (YCSB) has emerged as the most-widely used NoSQL benchmark for comparing NoSQL databases on create, read, update, delete (CRUD) and scan operations.

Vincent Reniers, Dimitri Van Landuyt, Ansar Rafique, and Wouter Joosen. 2017. ***On the State of NoSQL Benchmarks****. In Proceedings of the 8th ACM/SPEC on International Conference on Performance Engineering Companion (ICPE '17 Companion). Association for Computing Machinery, New York, NY, USA, 107–112. DOI:https://doi.org/10.1145/3053600.3053622*



Survey of existing NoSQL benchmarks

- Taking into account:
- (i) their supported database technologies,
- (ii) data models,
- (iii) the supported workload types.

- Discuss different design strategies and argue in favour of benchmark suites that encompass several specific application domains from graph stores to document stores, while still allowing for a general comparison between NoSQL databases in terms of common functionality.

Three properties for a relevant and widely applicable NoSQL benchmark

P1. Database support and applicability

- How widely applicable is the benchmark? Is the benchmark focused on a specific family of NoSQL databases (e.g. wide-column stores, graph stores), multiple families, or to any NoSQL database? Does the benchmark remain relevant for specific application scenarios (e.g. document search)?

Three properties for a relevant and widely applicable NoSQL benchmark

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P2. Data models

- Considering the large variety of data in Big Data applications and categories of data models in NoSQL, the benchmark has to generate various data sets that are aimed at testing the supported data models: graphs, regular key-value pairs, and even relational or connected and nested data.
- The generated data should be tailored for the purpose of evaluating the database and its model-associated functionality: e.g., by systematically varying the number of edges between nodes, varying the level of nesting depth in a JSON document, and the type of data used.
- In the ideal case, the user can supply the benchmark with sample data from a production environment, after which similar data is generated to allow relevant benchmarks for specific application scenarios.
- This property takes into account which data models are supported in the data generation step of the benchmark.

Three properties for a relevant and widely applicable NoSQL benchmark

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P3. Workloads

- Closely tied to P2 are the workloads which can be executed on the inserted data set. When no support is provided for e.g. graph generation, no graph traversing workloads will be possible either.
- The workloads should be able to fully exploit the capabilities of each NoSQL database, while allowing comparison on a subset of mutually supported functionality between a certain set of NoSQL databases.
- For example, the user provides a workload description of its search queries and applies this workload to its data modelled as a graph, and as a document, to identify which database and data model can potentially be faster for search.
- This property takes into account which types of workloads are supported in the benchmark.

THE STATE OF NOSQL BENCHMARKS

- Nowadays, there exist standardised organisations:
  - Transaction Processing Council (TPC) and the
  - Standard Performance Evaluation Corporation (SPEC)
- Release standardised benchmarks according to strict guidelines and review processes.
- Unfortunately, as of yet none of these organisations provide benchmarks for NoSQL databases.
- However, TPC does provide benchmarks for Big Data technologies such as TPC-H, TPC-DS, TPCx-HS for Hadoop or TPCx-BB for Hive and Spark with the possibility for structured, semi-structured and unstructured data generation.
- Yahoo! Cloud Serving Benchmark (YCSB), has become the de-facto standard for cross-family comparison of NoSQL DB.

www.tpc.org

The TPC is a non-profit corporation focused on developing data-centric benchmark standards and disseminating objective, verifiable data to the industry.

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Recent Benchmark Activities

- 02/02/26 Lenovo added a new **TPCx-AI result**: 893.92 AIUCpm@3 - 135.11 USD/AIUCpm - available: 2/2/2026
- 12/23/25 TTA added a new **TPCx-IoT result**: 16,969,033 IoTps - 30.18 USD/kIoTps - available: 12/23/2025
- 12/17/25 Timecho added a new **TPCx-IoT result**: 24,653,819 IoTps - 19.76 USD/kIoTps - available: 12/17/2025
- 11/17/25 HPE changed the availability date for a **TPC-H result**: 1,700,864 QphH@3,000GB - 216.31 USD/kQphH@3,000GB - available: 5/22/2026
- 11/17/25 HPE changed the availability date for a **TPC-H result**: 2,500,231 QphH@3,000GB - 247.03 USD/kQphH@3,000GB - available: 5/22/2026
- 11/17/25 HPE changed the availability date for a **TPC-H result**: 4,036,908 QphH@10,000GB - 332.00 USD/kQphH@10,000GB - available: 5/22/2026
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- 11/17/25 HPE added a new **TPC-H result**: 4,036,908 QphH@10,000GB - 332.00 USD/kQphH@10,000GB - available: 5/22/2026
- 10/01/25 HPE added a new **TPC-H result**: 1,531,016 QphH@1,000GB - 192.37 USD/kQphH@1,000GB - available: 10/1/2025
- 10/01/25 HPE added a new **TPC-H result**: 951,753 QphH@3,000GB - 241.17 USD/kQphH@3,000GB - available: 10/1/2025

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Key-value store benchmarks

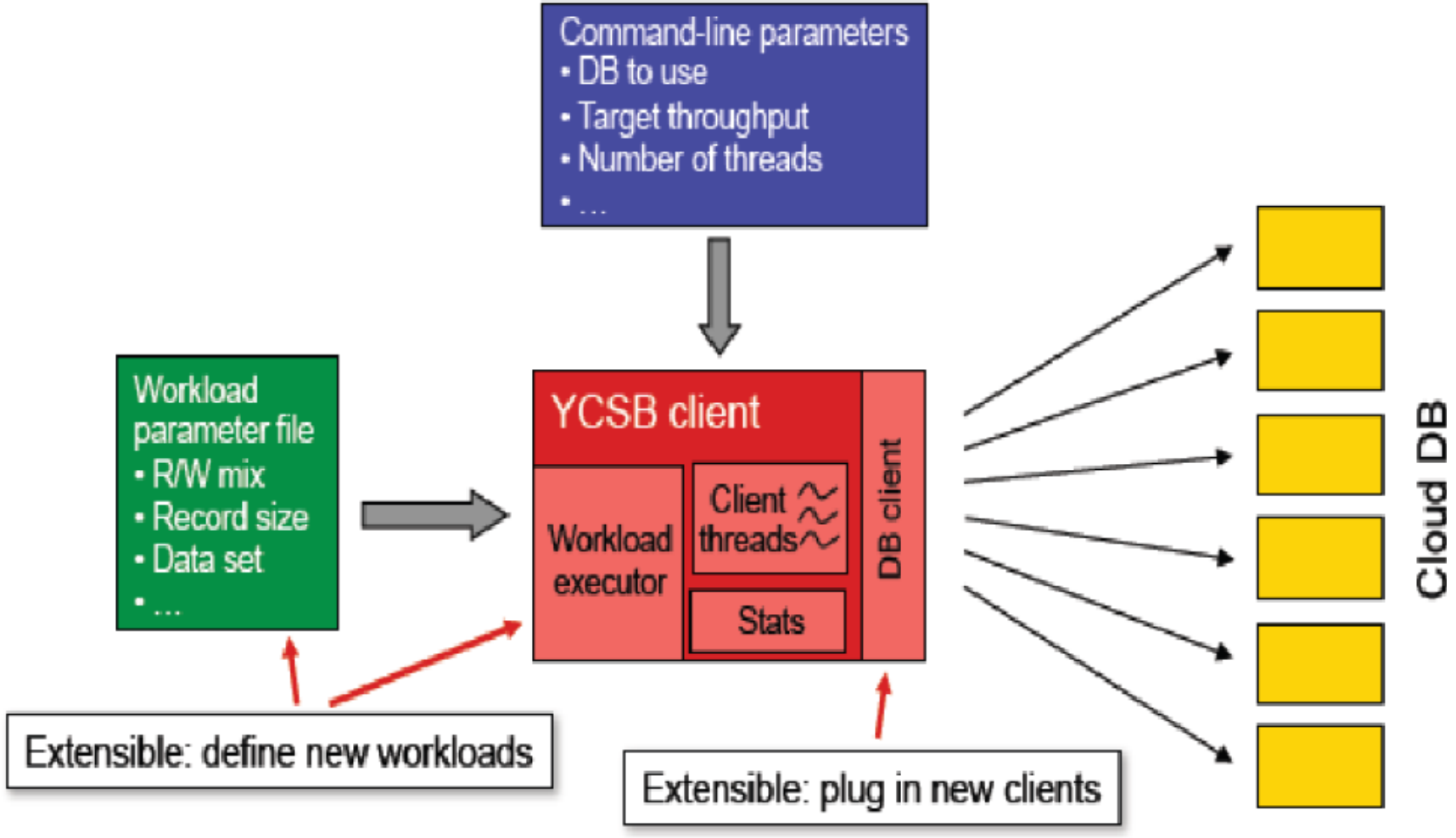
Yahoo! Cloud Serving Benchmark (YCSB)

- YCSB is able to execute workloads on read, write, update and scan operations on a primary key for various NoSQL databases such as Cassandra, MongoDB and Hbase.

KVZone

- Evaluates BerkeleyDB, Tokyo Cabinet, SQLite and Alphard.
- In contrast to YCSB, it has barely gained visibility, while YCSB has seen wide-spread adoption.

YCSB


Command-line parameters
• DB to use
• Target throughput
• Number of threads
• …
Workload parameter file
• R/W mix
• Record size
• Data set
• …
YCSB client
Workload executor
Client threads
Stats
DB client
Cloud DB
Extensible: define new workloads
Extensible: plug in new clients

YCSB Workloads

- Workload A: Update heavy workload. This workload has a mix of 50/50 reads and updates.
- Workload B: Read mostly workload. This workload has a 95/5 reads/update mix.
- Workload C: Read only. This workload is 100% read.
- Workload D: Read latest workload. In this workload, new records are inserted, and the most recently inserted records are the most popular.
- Workload E: Short ranges. In this workload, short ranges of records are queried, instead of individual records.
- Workload F: Read-modify-write. In this workload, the client will read a record, modify it, and write back the changes.

Document store benchmarks

NoWog

- It allows user-specifed workload descriptions on the basis of a generic grammar.
- The workload descriptions in an abstract grammar are then translated to database-specific operations, which results in CRUD operations executed on a set of supported database through mapping. However, the functionality is limited to CRUD operations.

SSB+

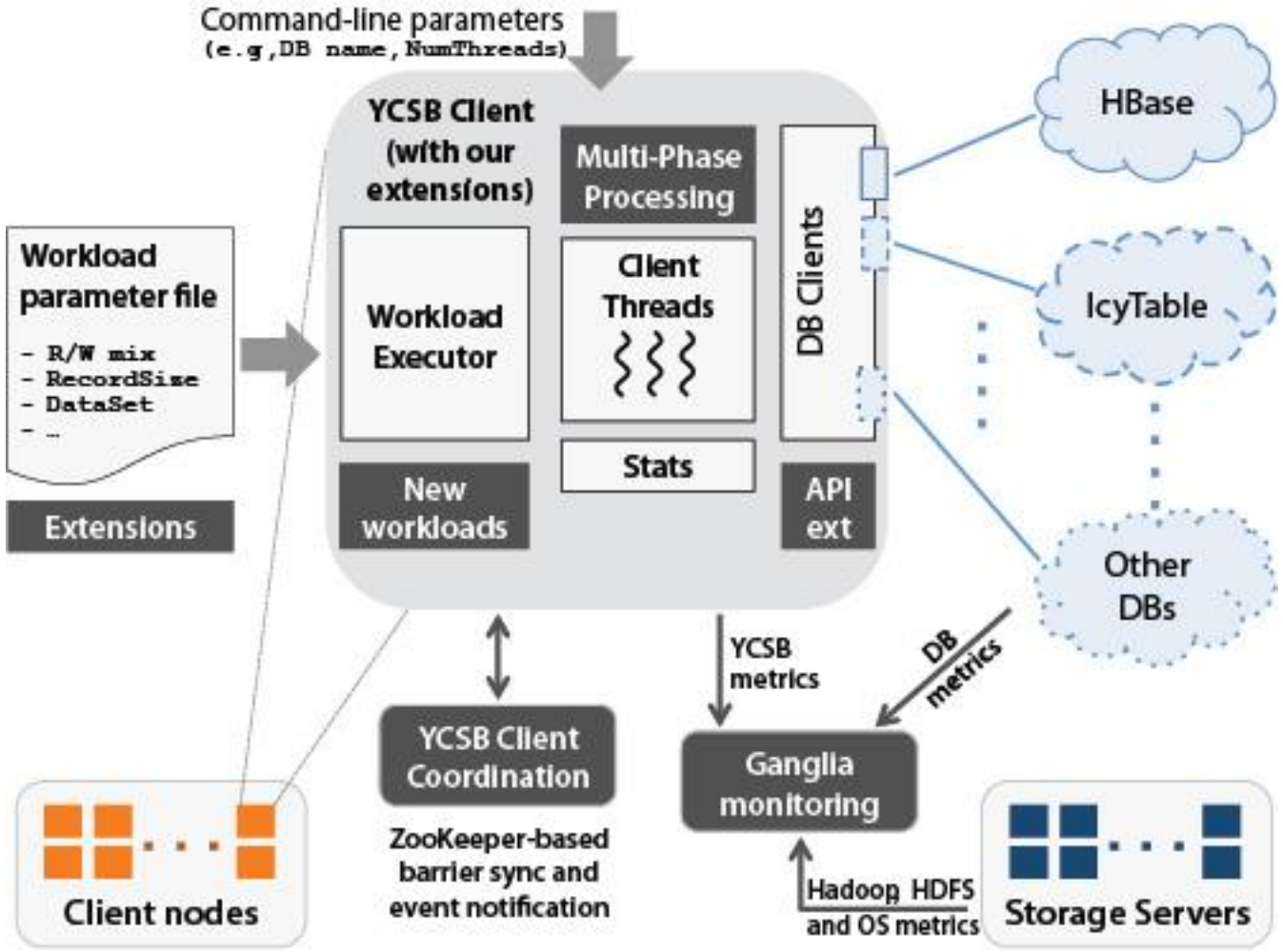
- SSB+, an extension on a decision support benchmark for relational databases, namely Star Schema Benchmark (SSB), and allow for the generated tables to be stored in a document and column-oriented databases.
- In addition, the OLAP SQL queries are translated to the respective NoSQL database.

Column store benchmarks

YCSB++

- YCSB++ is an extension on YCSB for benchmarking advanced features of scalable table stores (i.e. column-family stores) such as Apache HBase and Accumulo.
- The benchmark provides distributed synchronization between multiple benchmark clients and is able to measure eventual consistency, bulk loading, and the effect of optimizations for batch writing such as table pre-splitting.
- In addition, the benchmark can also measure the performance overhead of additional features such as access control and collects monitoring information on resource metrics at each cluster node.

YCSB ++



Source: https://www.pdl.cmu.edu/ycsb++/

Linked Data Benchmark Council (LDBC)

https://ldbcouncil.org

- The Linked Data Benchmark Council (LDBC) is a non-profit organization aiming to define standard graph benchmarks to foster a community around graph processing technologies. LDBC consists of members from both industry and academia, including organizations and individuals.
- LDBC currently offers the following benchmarks:
  - Graphalytics: Graph algorithms for graph analytical platforms
  - Financial Benchmark: Benchmark for financial workloads, targeting distributed systems
  - Semantic Publishing Benchmark: An RDF-based benchmark for semantic databases
  - Social Network Benchmark Suite (SNB): The SNB targets database management systems with graph-processing capabilities. It consists of two workloads, Interactive and Business Intelligence

Graph store benchmarks

GDB

- GDB is a distributed graph database benchmarking framework on operations such as exploring a node's neighborhood and finding the shortest path.

XGDBench

- XGDBench is an extension to YCSB for benchmarking graph stores by adding a graph data model and workload generator.

LinkBench

- LinkBench is a database benchmark based on social graph data from Facebook which mimics similar access patterns and operations.

Graph store benchmarks

GraphDB

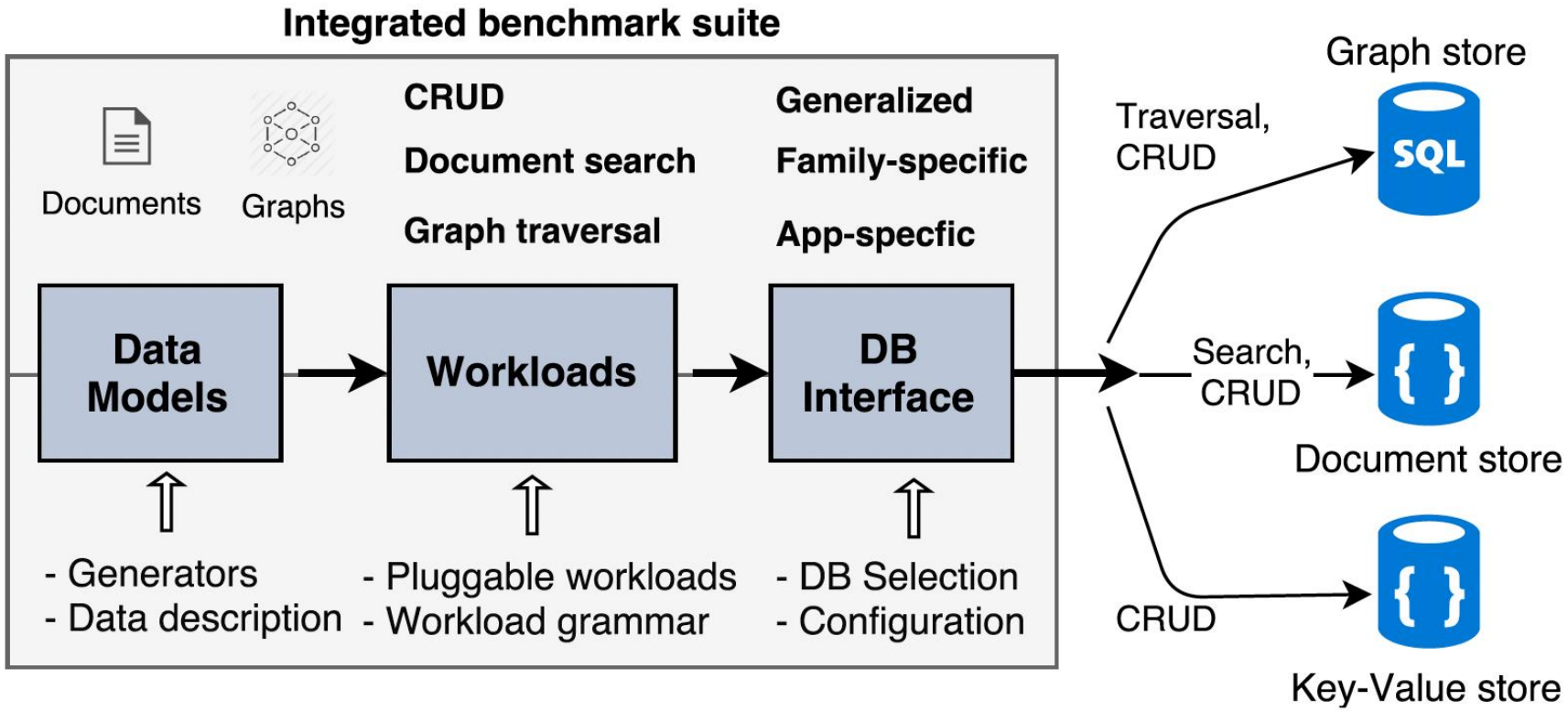
- The project graphdb-benchmarks is a benchmark between popular graph databases. Currently the framework supports Titan, OrientDB, Neo4j and Sparksee.
- The purpose of this benchmark is to examine the performance of each graph database in terms of execution time.
- The benchmark is composed of four workloads, Clustering, Massive Insertion, Single Insertion and Query Workload. Every workload has been designed to simulate common operations in graph database systems

NoSQL benchmarks

NoSQL benchmarks and the capabilities in data set generation and workloads

Applicability (P1)	Benchmarks	Data sets (P2)	Workloads (P3)
Key-value stores	YCSB [8], KVZone [13]	Key-value pairs of synthetic strings	CRUD + Range queries
Document stores	NoWog[22], Lungu et al.[18], SSB+[19]	Nested documents, limited generation options	Generic workload descriptions, CRUD, limited document queries
Column stores	YCSB++[28], Pirzadesh et al.[23], SSB+[19]	Structured data in tables with e.g. column families, limited generation options	CRUD, bulk loading, consistency evaluation, extensive range queries, limited table queries
Graph stores	Beis et al.[4], Ciglan et al.[7], GDB[16], XGDBench[9], LinkBench [2]	Multiple synthetic graph generators with e.g. varying clustering coefficients an edge connectivity preferences	Node and edge storage and retrieval, graph traversal, neighborhood discovery, clustering algorithms, ...

NoSQL benchmark suite proposal



Source: *Vincent Reniers, Dimitri Van Landuyt, Ansar Rafique, and Wouter Joosen. 2017. On the State of NoSQL Benchmarks. In Proceedings of the 8th ACM/SPEC on International Conference on Performance Engineering Companion (ICPE '17 Companion). Association for Computing Machinery, New York, NY, USA, 107–112.*

YCSB variations

- V. Razdan, A. Modi, S. Dumbare and R. Jobanputra, "**YCSB+T**: Benchmark for transactional databases and performance," *2017 International conference of Electronics, Communication and Aerospace Technology (ICECA)*, Coimbatore, India, 2017, pp. 406-409, doi: 10.1109/ICECA.2017.8212845.
- S. Sidhanta, S. Mukhopadhyay and W. Golab, "**Dyn-YCSB**: Benchmarking Adaptive Frameworks," *2019 IEEE World Congress on Services (SERVICES)*, Milan, Italy, 2019, pp. 392-393, doi: 10.1109/SERVICES.2019.00119.
  - https://github.com/ssidhanta/YCSBDEMOClient

Transactional YCSB

- Transactional YCSB: Benchmarking ACID-Compliant NoSQL Systems with Multi-Operation Transactions
  - Beckermann, Benedikt Maria (2025): Transactional YCSB: Benchmarking ACID-Compliant NoSQL Systems with Multi-Operation Transactions. Datenbanksysteme für Business, Technologie und Web (BTW 2025). DOI: 10.18420/BTW2025-67. Gesellschaft für Informatik, Bonn. EISSN: 2944-7682. pp. 1019-1030. Student Track. Bamberg. 3.-7. März 2025

  https://dl.gi.de/server/api/core/bitstreams/76335033-a076-4e4a-80f5-c4d2f1f756ed/content
- Transactional YCSB is an extension of the benchmarking framework YCSB that enables the evaluation of ACID-compliant NoSQL systems for workloads consisting of multi-operation transactions.
- https://github.com/greenBene/transactional-ycsb

Conclusion

- The majority of benchmarks are complementary initiatives that are aimed at specific NoSQL technologies or technology families, concrete data models or specific application workloads.
- We identified that such efforts have seen the largest development in graph stores concerning the generation of synthetic graphs and various workloads.
- However, document and column-store benchmarks are specifically still lacking in these capabilities.
- We conclude that cross-family NoSQL benchmark initiatives are currently limited to YCSB, although the question of storing graph data in a document store, RDBMs or graph store can be a relevant question. However, the tools and means to evaluate and aid in this decision making are currently non-existent.

Examples of practical works

- *NoSQL Databases: MongoDB vs Cassandra. Veronika Abramova and Jorge Bernardino. 2013. In Proceedings of the International C* Conference on Computer Science and Software Engineering (C3S2E '13). Association for Computing Machinery, New York, NY, USA, 14–22. DOI:https://doi.org/10.1145/2494444.2494447*

  https://dl.acm.org/doi/abs/10.1145/2494444.2494447



Examples of practical works

- *Performance Evaluation of NoSQL Document Databases: Couchbase, CouchDB, and MongoDB*. Carvalho, Inês, Filipe Sá, and Jorge Bernardino. 2023. Algorithms 16, no. 2: 78. https://doi.org/10.3390/a16020078
- https://www.mdpi.com/1999-4893/16/2/78

- *NewSQL Databases Assessment: CockroachDB, MariaDB Xpand, and VoltDB*. Pina, Eduardo, Filipe Sá, and Jorge Bernardino. 2023. Future Internet 15, no. 1: 10. https://doi.org/10.3390/fi15010010
- https://www.mdpi.com/1999-5903/15/1/10

Examples of practical works

- *Experimental Evaluation of Graph Databases: JanusGraph, Nebula Graph, Neo4j, and TigerGraph*. *Monteiro, Jéssica, Filipe Sá, and Jorge Bernardino. 2023. Applied Sciences 13, no. 9: 5770. https://doi.org/10.3390/app13095770*
- ***https://www.mdpi.com/2076-3417/13/9/5770***

- *Deep Learning for Big Data. Filipe Correia, Ana Madureira, Jorge Bernardino. 2022. IBICA 2021. Lecture Notes in Networks and Systems, vol 419. Springer, Cham.*
- ***https://doi.org/10.1007/978-3-030-96299-9_20***


QUESTIONS?
COMMENTS?
CONCERNS?

YCSB

- https://benchant.com/blog/ycsb
- https://benchant.com/blog/ycsb

- https://benchant.com/blog/ycsb
- https://benchant.com/blog/database-benchmarking
- https://benchant.com/ranking/database-ranking

YCSB

https://github.com/brianfrankcooper/YCSB/wiki/Core-Workloads#running-the-workloads

Running the workloads

All six workloads have a data set which is similar. Workloads D and E insert records during the test run. Thus, to keep the database size consistent, we recommend the following sequence:

1. Load the database, using workload A's parameter file (workloads/workloada) and the "-load" switch to the client.
2. Run workload A (using workloads/workloada and "-t") for a variety of throughputs.
3. Run workload B (using workloads/workloadb and "-t") for a variety of throughputs.
4. Run workload C (using workloads/workloadc and "-t") for a variety of throughputs.
5. Run workload F (using workloads/workloadf and "-t") for a variety of throughputs.
6. Run workload D (using workloads/workloadd and "-t") for a variety of throughputs. This workload inserts records, increasing the size of the database.
7. Delete the data in the database.
8. Reload the database, using workload E's parameter file (workloads/workloade) and the "-load switch to the client.
9. Run workload E (using workloads/workloade and "-t") for a variety of throughputs. This workload inserts records, increasing the size of the database.


End of part one
To Be Continued
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